



Core User Workflows

Workflow 1: AI-Powered Daily Check-in

1. User lands on `/dashboard` and clicks the pill-shaped "Log Mood" action bar.
2. Frontend posts mood and behavioral variables to FastAPI backend.
3. Backend calculates updated features and runs the XGBoost model to get `stress_score` and SHAP drivers.
4. Backend triggers the LangGraph pipeline (Component 2) which returns structured JSON insights.
5. UI instantly updates the dashboard Insight Cards with the new clinical synthesis.

Workflow 2: Conversational Mentorship (Aura)

1. User navigates to `/aura-chat` and sends a message.
2. FastAPI backend receives the message and embeds it using `mxbai-embed-large`.
3. Backend queries Supabase `pgvector` to retrieve the top 3 similar past memories.
4. Context + New Message are injected into LLM-II (Claude/Phi-4).
5. Aura responds empathetically. The new conversation chunk is asynchronously embedded and saved to the vector DB for future recall.

Workflow 3: RLHF Smart Nudges

1. Backend cron job runs daily, querying user schedule and current ML stress score.
2. LLM-III references the `user_nudge_feedback` table for past successful nudge types.
3. Generates a specific nudge (e.g., "5-min breathing before your 3 PM meeting").
4. Frontend displays nudge. If user clicks "Start", it logs +1 feedback. If dismissed, -1 feedback.